

Justify all answers. **Answers without sufficient justification will receive no credit. A numeric answer alone, without explanation and supporting work, is not acceptable.** Submit as a pdf to canvas.

1. (20 points) Use existential and universal quantifiers to express the statement “Every grandparent is a parent of a parent” using the propositional functions $G(g, y)$, which represents “ g is a grandparent of y ,” and $P(x, y)$, which represents “ x is a parent of y .”
2. (20 points) Let A and B be sets. Prove that if $A \subseteq B$, then $A \cap B = A$.
3. (20 points) Use mathematical induction to show that

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1},$$

whenever n is a positive integer.

4. (20 points) Solve the recurrence relation together with the initial conditions:

$$a_n = 4a_{n-1} - 4a_{n-2} + 3^n \quad \text{for } n \geq 2, \quad a_0 = 6, \quad a_1 = 8.$$

5. (20 points) You have n identical balloons to give to three children: Alice, Ben, and Chloe.
 - Alice can receive **any nonnegative** number of balloons.
 - Ben can receive **at most 2** balloons;
 - Chloe must receive an **even** number of balloons;

How many ways are there to distribute the balloons?

6. (20 points) A *bitonic array* $A[1..n]$ is one that strictly increases up to a (unique) peak index p and then strictly decreases; that is, $A[1] < A[2] < \cdots < A[p] > A[p+1] > \cdots > A[n]$ for some $1 \leq p \leq n$. Find a recurrence relation that describes the running time of the following algorithm: Find the peak index in a bitonic array of n numbers by performing a binary-search-style procedure that compares the middle element $A[\lfloor n/2 \rfloor]$ with its right neighbor $A[\lfloor n/2 \rfloor + 1]$ to decide whether to recurse on the left half or the right half, repeating until a single index remains. Then apply the Master Theorem to obtain the big- O estimate for the running time of the algorithm.